

Communication

Date	03 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Communication Plan:

Effective communication was essential for the successful development and demonstration of the Smart Lender project. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed while building the ML-powered loan approval prediction system.

S. No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Google Meet / WhatsApp Group	Development Team	Discuss daily progress, tasks, and blockers
2	Progress Update	Weekly	MS Teams / Email	Team Members & Mentor	Share weekly progress report and next steps
3	Issue / Bug Discussion	As Needed	WhatsApp / Slack	Development Team	Resolve technical issues, data or model errors
4	Stakeholder Review	Bi-Weekly	Zoom / MS Teams	Team, Mentor & Stakeholders	Review project milestones and gather feedback
5	Final Demo Rehearsal	Once	Google Meet	Entire Team	Rehearse final project demonstration flow
6	Model Evaluation Review	Once	MS Teams	Development Team & Mentor	Compare Decision Tree, Random Forest, KNN and XGBoost results to finalize best model

Communication Challenges & Resolutions:

S. No	Challenge Faced	Resolution / Action Taken
1	Initial Random Forest model (default parameters) showed severe overfitting — 100% training accuracy but only 68.7% testing accuracy	Flagged during an issue discussion session; tuned hyperparameters (max_depth=10, min_samples_split=5, min_samples_leaf=2) which reduced overfitting and raised testing accuracy to 70.2%
2	Manual hyperparameter tuning still left accuracy suboptimal with a noticeable train-test gap	Team adopted GridSearchCV with 5-fold cross-validation across n_estimators, max_depth, min_samples_split and min_samples_leaf to systematically find the best combination, improving testing accuracy to 70.6% with a narrower train-test gap
3	Converting multiple categorical features (Gender, Married, Dependents, Education, Self_Employed, Property_Area, Loan_Status) into numeric form for model training	Used scikit-learn's LabelEncoder; team reviewed the encoding logic together during standup to ensure consistent transformation across all categorical columns before training